

NoSQL Databases types

Why are RDBMS not suitable for Big Data

- RDBMSs assume that data are
 - Dense
 - Largely uniform (structured data)
- Data coming from Internet are
 - Massive and sparse
 - Semi-structured or unstructured
- With massive sparse data sets, the typical storage mechanisms and access methods get stretched

NoSQL Definition

Next Generation Databases mostly addressing some of the points: being **non-relational**, **distributed**, **open-source** and **horizontal scalable**. The original intention has been **modern web-scale databases**. The movement began early 2009 and is growing rapidly. Often more characteristics apply as: **schema-free**, **easy replication support**, **simple API**, **eventually consistent / BASE** (not ACID), a **huge data amount**, and more.

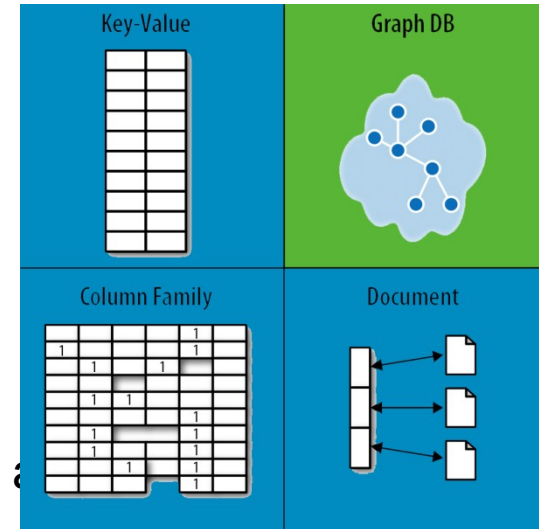
NoSQL

- NoSQL stands for:
 - No Relational
 - No RDBMS
 - Not Only SQL
- NoSQL is an umbrella term for all databases and data stores that don't follow the RDBMS principles
 - A class of products
 - A collection of several (related) concepts about data storage and manipulation
 - Often related to large data sets

NoSQL Database Types

Discussing NoSQL databases is complicated because there are a variety of types:

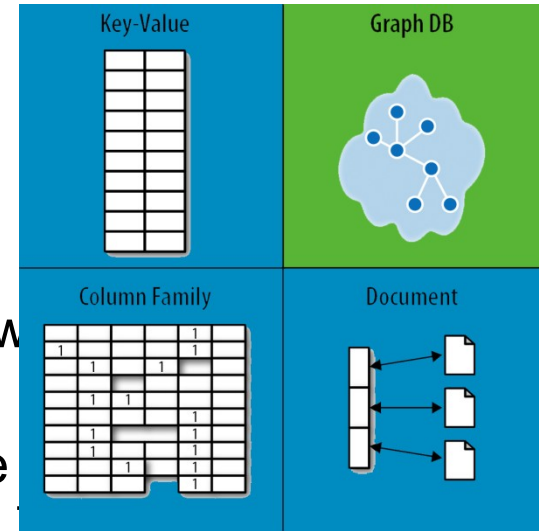
- Sorted ordered Column Store
 - Optimized for queries over large datasets, and store columns of data together, instead of rows
- Document databases:
 - pair each key with a complex data structure known as a document
- Key-Value Store :
 - are the simplest NoSQL databases. Every single item in the database is stored as an attribute name (or 'key'), together with its value.
- Graph Databases :
 - are used to store information about networks of data, such as social connections.



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Document Databases (Document Store)

- Documents
 - Loosely structured sets of key/value pairs in documents, e.g., XML, JSON, BSON
 - Encapsulate and encode data in some standard formats or encodings
 - Are addressed in the database via a unique key
 - Documents are treated as a whole, avoiding splitting a document into its constituent name/value pairs
- Allow documents retrieving by keys or contents
- Notable for:
 - MongoDB (used in FourSquare, Github, and more)
 - CouchDB (used in Apple, BBC, Canonical, Cern, and more)

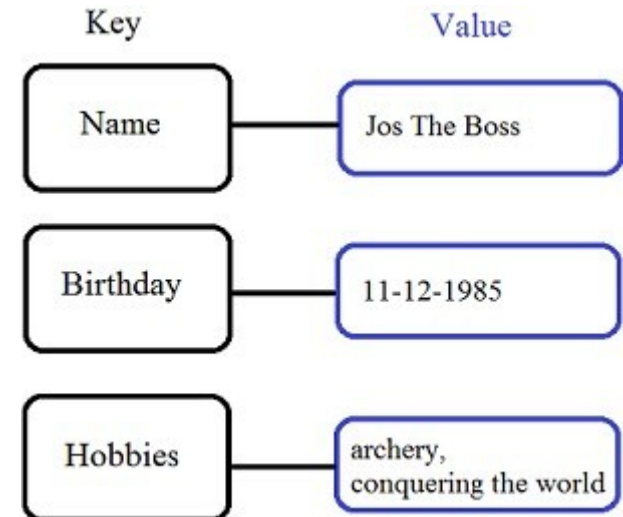
Document Databases (Document Store)

- The central concept is the notion of a "document" which corresponds to a row in RDBMS.
- A document comes in some standard formats like JSON (BSON).
- Documents are addressed in the database via a unique *key* that represents that document.
- The database offers an API or query language that retrieves documents based on their contents.
- Documents are schema free, i.e., different documents can have structures and schema that differ from one another. (An RDBMS requires that each row contain the same columns.)

Document Databases, JSON

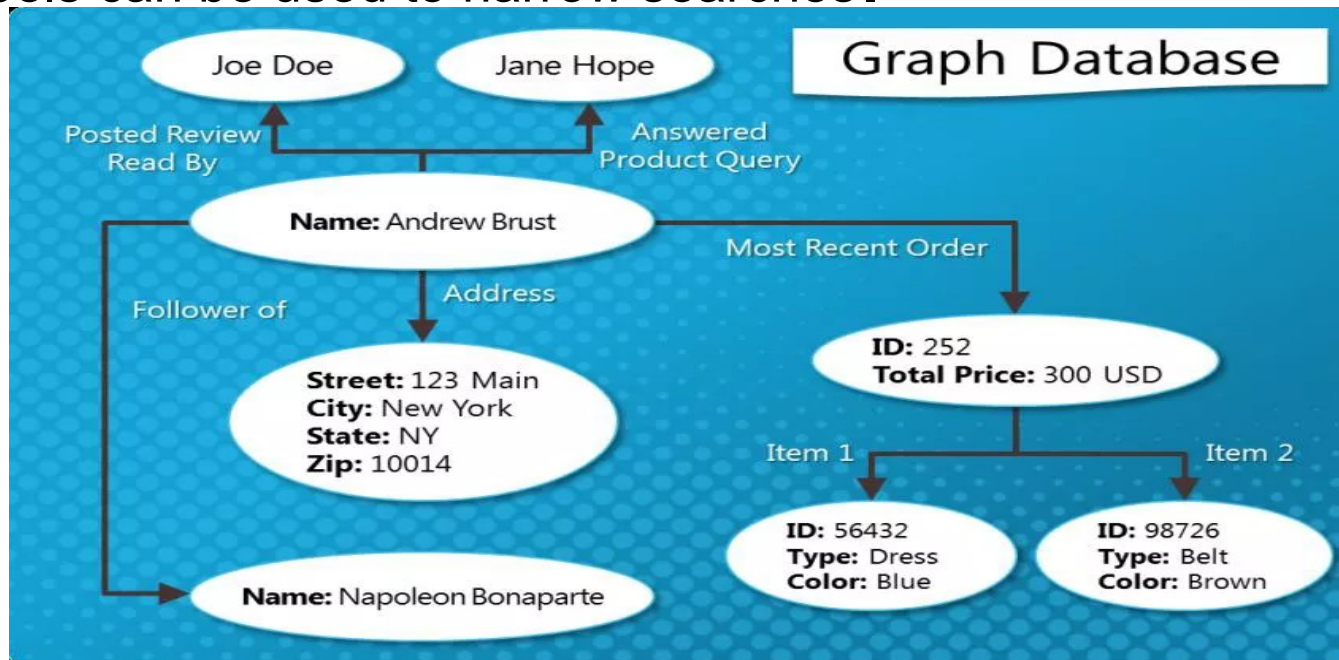
Key/Value stores

- Store data in a schema-less way
- Store data as maps
 - HashMaps or associative arrays
 - Provide a very efficient average running time algorithm for accessing data
- Notable for:
 - Couchbase (Zynga, Vimeo, NAVTEQ, ...)
 - Redis (Craiglist, Instagram, StackOverflow, flickr, ...)
 - Amazon Dynamo (Amazon, Elsevier, IMDb, ...)
 - Apache Cassandra (Facebook, Digg, Reddit, Twitter,...)
 - Voldemort (LinkedIn, eBay, ...)
 - Riak (Github, Comcast, Mochi, ...)



Graph Databases

- Graph-oriented
- Everything is stored as an edge, a node or an attribute.
- Each node and edge can have any number of attributes.
- Both the nodes and edges can be labelled.
- Labels can be used to narrow searches.



Dealing with Big Data and Scalability

- Issues with scaling up when the dataset is just too big
- RDBMS were not designed to be distributed
- Traditional DBMSs are best designed to run well on a “single” machine
 - _ Larger volumes of data/operations requires to upgrade the server with faster CPUs or more memory known as ‘scaling up’ or ‘Vertical scaling’
- NoSQL solutions are designed to run on clusters or multi-node database solutions
 - _ Larger volumes of data/operations requires to add more machines to the cluster, Known as ‘scaling out’ or ‘horizontal scaling’
 - _ Different approaches include:
 - Master-slave
 - Sharding (partitioning)

NoSQL Products/Projects

122 NoSQL Databases

- Cassandra
- CouchDB
- Hadoop & Hbase
- MongoDB
- StupidDB
- Etc.

NoSQL Distinguishing Characteristics

- Large data volumes
 - Google's "big data"
- Scalable replication and distribution
 - Potentially thousands of machines
 - Potentially distributed around the world
- Queries need to return answers quickly
- Mostly query, few updates
- Asynchronous Inserts & Updates
- Schema-less
- ACID transaction properties are not needed – BASE
- CAP Theorem
- Open source development

Types of Next Generation Databases(SUMMARY)

BASE Transactions

- Acronym contrived to be the opposite of ACID
 - **B**asically **A**vailable,
 - **S**oft state,
 - **E**ventually Consistent
- Characteristics
 - Weak consistency – stale data OK
 - Availability first
 - Best effort
 - Approximate answers OK
 - Aggressive (optimistic)
 - Simpler and faster

Brewer's CAP Theorem

A distributed system can support only two of the following characteristics:

- Consistency
- Availability
- Partition tolerance

The slides from Brewer's July 2000 talk do not define these characteristics.

Consistency

- all nodes see the same data at the same time
- client perceives that a set of operations has occurred all at once More like Atomic in ACID transaction properties

Availability

- node failures do not prevent survivors from continuing to operate ,Every operation must terminate in an intended response

Partition Tolerance

- the system continues to operate despite arbitrary message loss , Operations will complete, even if individual components are unavailable.

Column Store Comments

- More efficient than row (or document) store if:
 - Multiple row/record/documents are inserted at the same time so updates of column blocks can be aggregated
 - Retrievals access only some of the columns in a row/record/document

NoSQL Summary

- NoSQL databases reject:
 - Overhead of ACID transactions
 - “Complexity” of SQL
 - Burden of up-front schema design
 - Declarative query expression
 - Yesterday’s technology
- Programmer responsible for
 - Step-by-step procedural language
 - Navigating access path

Summary

- SQL Databases
 - Predefined Schema
 - Standard definition and interface language
 - Tight consistency
 - Well defined semantics
- NoSQL Database
 - No predefined Schema
 - Per-product definition and interface language
 - Getting an answer quickly is more important than getting a correct answer

Web References

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